MediScope

Raw patient data, made actionable.

Healthcare Analytics

MIMIC-III ICU · DE-IDENTIFIED

Tools: Python · Pandas · Tableau

Institute: Newton School of Technology

TEAM G-17 · SECTION D

Satyam Kumar
Project Lead

Sambhav Kumar
Visualization

Shagun Singh
Statistical Analysis

Yash
Documentation

Harsh
Research & QA

Mentor: Archit Raj · April 2026

WHY DOES THIS PROJECT MATTER?

Context & Problem Statement

ICU operations are highly resource-intensive.

Small inefficiencies quickly add up to lost time, cost, and capacity.

1

Unpredictable Patient Volumes

91% of admissions are emergency cases.

Forecasting becomes difficult.

2

Laboratory Inefficiency

More than 760 tests per patient.

Abnormal results stay flat even during spikes.

3

No Data-Driven Visibility

Leadership lacks a clear view of cost drivers.

Department, admission type, and patient segment insights are fragmented.

MediScope closes the visibility gap.

It turns raw ICU records into structured decision support.

SOURCE · SIZE · CLEANING · ETL PIPELINE

Data Engineering

Dataset Overview

Source	MIMIC-III Clinical Database (de-id)
Records	76,069 laboratory event rows
Patients	100 unique ICU patients
Admissions	129 total hospital stays
Tables Joined	LABEVENTS, ADMISSIONS, PATIENTS, D_LABITEMS
Key Variables	subject_id, hadm_id, itemid, is_abnormal, los_days, admission_type, gender
Output File	clean_lab_master_v3.csv

99.99% of records preserved after cleaning

Python ETL Pipeline

01

Extract

- 5 raw CSV files ingested
- Profiled types, nulls, and cardinality

02

Transform

- Removed 5 unresolvable null records
- Built `los_days` and `is_abnormal`
- Merged on `subject_id` + `hadm_id`

03

Load

- Exported `clean_lab_master_v3.csv`
- Built Tableau-ready subsets
- Optimized for fast queries

Key Insights from Exploratory Data Analysis

Business-language findings from four core visualisations

1	2	3
Emergency Dominance	**7.6-Day LOS Gap**	**Gender Alert Disparity**
• 91% of patients arrive via Emergency • Average stay: 28.8 days • 36% longer than elective cases	• Emergency stays outlast elective stays • Gap signals avoidable operational drag • Potentially displaces elective revenue	• Female patients trigger more abnormal flags • Emergency: 41.5% vs 37.2% • Likely reference-range miscalibration

Advanced Analysis

Statistical Validation · Segmentation · Root Cause

1

Independent T-Test

- **Groups:** Emergency vs. Elective LOS
- **Mean diff:** +7.6 days (+36%)
- **p-value:** < 0.0001 ✓
- **Takeaway:** The LOS gap is real.

2

Chi-Square Test

- **Groups:** Gender vs. Abnormal Flag
- **Finding:** Female > Male across all types
- **p-value:** < 0.0001 ✓
- **Takeaway:** Reference ranges likely drive the disparity.

3

K-Means Clustering (k=3)

- **Cluster 0 (n=60):** Standard acuity
- **Cluster 1 (n=26):** Elevated risk
- **Cluster 2 (n=14):** Critical outliers
- **Impact:** 14 patients consume disproportionate resources.

⚠ **Charts not available:** Dept. Quadrant (Volume vs. Abnormal Rate)

Pattern: Volume spikes ↗ while abnormal rate stays flat ≈39%

Signal: Possible over-testing

KPI & Metrics Framework

Core operational metrics from 76,069 lab records

76,069

Total Lab Records

100

Unique ICU Patients

129

Total Admissions

39.1%

Overall Abnormal Rate

22.5d

Average LOS

760.7

Tests per Patient

What the metrics show

- Abnormal rate: 39.1%
- Average LOS: 22.5 days
- Tests per patient: 760.7
- Emergency share: 91%

Key interpretation

- ICU acuity is high
- Reference ranges may be over-alerting
- Testing volume suggests standing orders
- Reactive mode limits capacity buffer

Operational insight The 36% LOS gap between Emergency and Elective patients suggests uneven flow and higher complexity.

Dashboard Walkthrough

Three-tier Tableau decision-support framework

Dashboard 1 — Macro KPIs & Volume Overview

Top Row — Macro KPIs

- Total records
- Abnormal rate
- Unique patients
- Avg LOS
- Blood specimen mix
- Tests per patient

Middle Row — Volume & Distribution

- Lab tests by category
- Hematology leads: 33,567
- Normal / Abnormal / Delta split
- LOS distribution

Bottom Row — Time Series

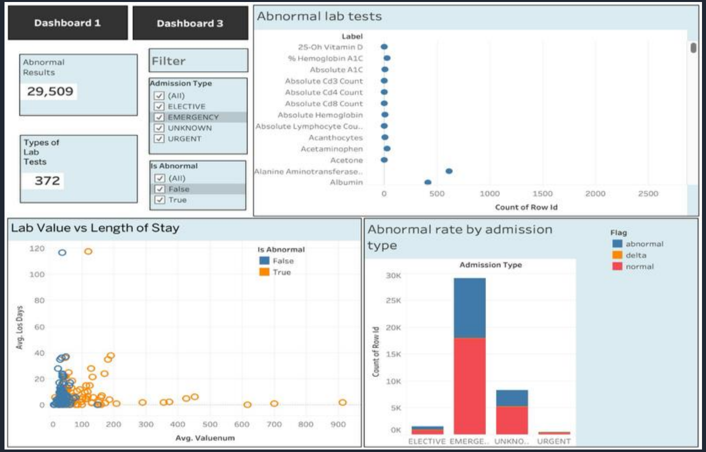
- Monthly volume spikes
- Abnormal rate stays flat
- Clear over-testing signal

⚠ Volume surges. Abnormal rate stays flat. Over-testing signal.

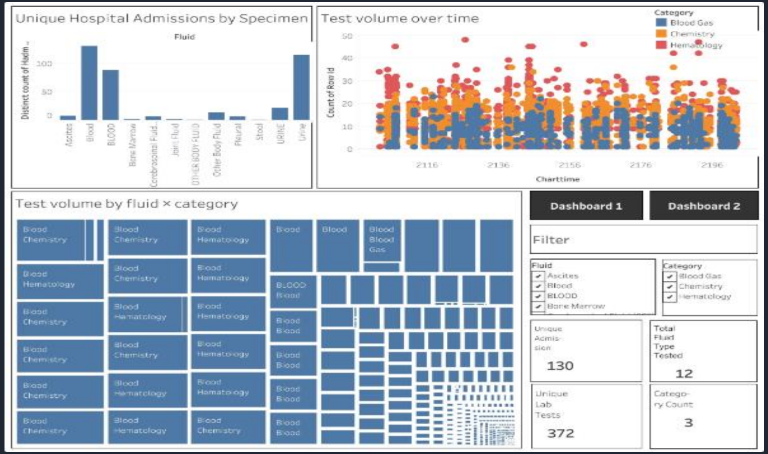
🔗 public.tableau.com/app/profile/sambhav.kumar2781/viz/dv_p7/Dashboard1

Dashboard Walkthrough — Clinical & Specimen Views

Dashboards 2 & 3 focus on admission-stratified risk and fluid-level detail.



Dashboard 2 — Admission-Stratified Clinical Risk



Dashboard 3 — Specimen & Test Volume Distribution

Abnormal tests

Grouped by test label.

Lab value vs LOS

Scatter shows clinical spread.

Admission types

Stacked bars compare mix.

Admissions by fluid

Specimen-level breakdown.

Test volume over time

Trends by category.

Fluid × category

Treemap highlights combinations.

Clinical focus: abnormal tests, LOS, and admission type.

Specimen focus: fluid mix, volume trends, and category combinations.

Actionable Recommendations

Three high-impact interventions tied directly to the findings

1 Triage Optimisation

Root cause: Emergency patients stay 36% longer

- Fast-track discharge for non-critical emergency admits
- Use a dedicated rapid-assessment team
- Run daily LOS review rounds from day 18–25

10% LOS reduction → 2.9 bed-days saved per patient

264 recovered bed-days per 100-patient cycle

2 Gender-Specific Lab Reference Ranges

Root cause: Female patients show higher abnormal flag rates

- Audit the D_LABITEMS reference range table
- Recalibrate Hematology & Chemistry thresholds
- Run a 30-day parallel validation

5–10% fewer false-positive alerts

~2,083 fewer unnecessary physician reviews per 100-patient cycle

3 Hematology EHR Hard Stops

Root cause: Highest volume, highest abnormal rate, and repeated spikes

- Add mandatory justification for repeat Hematology orders within 24 hours
- Allow physician override with a documented indication
- Review monthly peer-benchmark dashboards

Projected 15% reduction in unnecessary test spikes

Lower reagent use, equipment wear, and technician overtime

Impact & Value Estimation

Projected outcomes per 100-patient cycle

Key Metrics

15

Test Spike Reduction

Repeat-test reduction

2,083

Physician Reviews Saved

Per 100-patient cycle

264

Bed-days Recovered

Per 100-patient cycle

Projected Outcomes

Recommendation	Lever	Outcome
Triage Optimisation	10% LOS ↓	264 bed-days recovered
Gender Ref. Ranges	7% false-positive ↓	~2,083 fewer reviews
Hematology Hard Stops	15% repeat-test ↓	Reagent + labour savings
Predictive LOS Model*	Early Cluster 2 ID	2–3× cost saving vs reactive escalation

Phase 2 note

Cluster 2 LOS model depends on expansion to 1,000+ patients.

ICU Cost Context

India: ₹15,000–₹50,000/bed-day

USA: $4,000+/bed-day

264 recovered bed-days × ₹30,000 avg ≈ **₹79.2 lakh** per cycle

Directional estimate pending billing data integration

Limitations & Next Steps

Constraints acknowledged · Phase 2 roadmap

⚠ Current Limitations

Sample Size (n=100)

Statistically significant.

Too small for robust ML training.

Limited for multi-hospital use.

No Direct Cost Data

Financial estimates use published benchmarks.

Directional only, not CFO-ready.

Single Institution

MIMIC-III reflects one tertiary care hospital.

Results may vary by region, season, and case mix.

No Billing Integration

Precise cost-benefit analysis needs linked billing codes.

Planned for Phase 2.

🚀 Phase 2 Roadmap

▶ Dataset Expansion

Scale to 1,000+ patients.

Enable stronger ML training and subgroup analysis.

▶ Financial Integration

Link billing and cost-accounting data to test IDs.

Replace benchmarks with precise estimates.

▶ Predictive LOS Model

Use K-Means clusters as labels.

Start with logistic regression, then gradient boosting.

Flag Cluster 2 risk on admission day.

▶ Real-time Tableau API

Connect to live EHR feeds.

Shift from retrospective review to proactive alerts.

Warn before 25-day breach.

Thank You

Questions & Discussion

Dataset

MIMIC-III Clinical Database De-identified · MIT Lab

Project Stack

Python · Pandas
Tableau Public

GitHub

github.com/Satyamkumar2610/SectionD_G-17_MediScope

Team G-17 · Section D

Newton School of Technology · April 2026

Mentor: Archit Raj